Chess.com Engineering Internship - AI/ML Take-Home Assignment

The task

Using Chess.com's Published Data API, choose **one** of the following two tracks that best matches your strengths:

> **Option A — Machine Learning: Predict Chess Game Outcomes**
> Build a model that predicts the outcome of a chess game (win, loss, or draw for White) using only pre-game information.
> Collect your dataset from recent Titled Tuesday tournaments (for example, Feb 10, 2026 and Mar 10, 2026). Engineer pre-game features, train a model, and evaluate its performance.

> **Option B — Applied AI / GenAI: Build an AI Chess Coach**
> Build a self-contained AI chess coach web application powered by an LLM and Chess.com's public API. For example, fetch a player's recent games or stats and help them understand their mistakes, analyze specific games, or get personalized study advice.

What we are looking for

Showcase your best AI/ML and engineering capabilities.

- **For Option A:** We value a simple, well-reasoned pipeline (data collection, feature selection without data leakage, appropriate train/test split, and honest evaluation) over a complex, leaderboard-topping model.
- **For Option B:** We value a working, useful core experience, thoughtful prompt/context engineering, and grounding the AI's responses in real game data over covering every possible feature.

Time

We don't expect you to spend more than 4 hours on this.
Please let us know roughly how much time you spent and how it was split between the parts of the project.

AI usage

You are welcome to use AI tools and agents.
Please share any agent transcripts together with your result.

What to submit

- A link to a Git repository with your code or notebooks (public, or shared with us)
- A README explaining how to run the project or reproduce your results locally, ideally with a single command

- A short note on your main decisions and trade-offs (for Option A: your data split, feature choices, and honest view of model quality; for Option B: your system design, context/prompting strategy, and limitations), and anything you would do next with more time
- Your time breakdown and any agent transcripts

Good luck, and have fun building!